

Rajalakshmi Engineering College

Name: Harsan J
Email: 240701179@rajalakshmi.edu.in
Roll no: 240701179
Phone: 8925466195
Branch: REC
Department: CSE - Section 10
Batch: 2028
Degree: B.E - CSE

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2024_28_III_OOPS Using Java Lab

2028_REC_OOPS using Java_Week 1_Q1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Gloria is responsible for monitoring the performance of two machines in a factory. She needs to determine which of the two machines is operating closest to the optimal temperature of 100 degrees Celsius using the relational operator.

Assist Gloria in displaying the machine's temperature, which is closer to 100, and the difference from 100.

Input Format

The first line of input consists of an integer N, representing the temperature of the first machine.

The second line consists of an integer M, representing the temperature of the second machine.

Output Format

The output prints "The integer closer to 100 is X with a difference of Y" where X is the temperature of the closer machine and Y is the difference from 100.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 90

80

Output: The integer closer to 100 is 90 with a difference of 10

Answer

```
// You are using Java
import java.util.Scanner;
class MachineTemperature {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int N = scanner.nextInt();
        int M = scanner.nextInt();

        int diffN = Math.abs(100 - N);
        int diffM = Math.abs(100 - M);

        if (diffN < diffM) {
            System.out.println("The integer closer to 100 is " + N + " with a difference
of " + diffN);
        } else if (diffM < diffN) {
            System.out.println("The integer closer to 100 is " + M + " with a difference
of " + diffM);
        } else {
            System.out.println("Both integers are equally close to 100 with a
difference of " + diffN);
        }
    }
}
```

Status : Correct

Marks : 10/10